

Fourier Transform in Seismology and Earthquake Engineering

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1 The Fourier Transform

Any function $f(t)$ can be decomposed into the sum or integral of harmonic waves of different frequencies, this is known as the Fourier series, and can be either real or complex.

The complex Fourier series is given by ;

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi t}{L}} \quad (1)$$

Where c_n are the complex Fourier coefficients and given by

$$c_n = \frac{1}{2L} \int_{-L}^L f(t) e^{-\frac{in\pi t}{L}} dt \quad (2)$$

Replacing c_n in equation (1) yields:

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{1}{2L} \int_{-L}^L f(t) e^{-\frac{in\pi t}{L}} e^{\frac{in\pi t}{L}} dt \quad (3)$$

We shall now define a number of parameters. T is the period and is given by $2L$, and ω is the angular frequency given by:

$$\omega = \frac{2\pi n}{T} \quad (4)$$

But $T = 2L$, simplifying equation (4) to:

$$\omega = \frac{n\pi}{L} \quad (5)$$

There are n frequencies as observed, and it is clear that the smallest change in frequency will be $\frac{\pi}{L}$, and therefore:

$$\Delta\omega = \frac{\pi}{L} \quad (6)$$

Equation 5 is of much importance to us as it changes the parameters of our function to frequency.

Replacing ω and $\Delta\omega$ in equation (3):

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{\Delta\omega}{2\pi} \int_{-L}^L f(t) e^{-i\omega t} e^{i\omega t} dt \quad (7)$$

Letting L approach infinity:

$$\lim_{L \rightarrow \infty} \Delta\omega = \delta\omega$$

Substituting in equation (7):

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt e^{i\omega t} d\omega \quad (8)$$

This is a Riemann sum, and therefore:

$$f(t) = \int_{-\infty}^{\infty} \frac{1}{2\pi} \left(\int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt \right) e^{i\omega t} d\omega \quad (9)$$

This (in brackets) is the Fourier transform of $f(t)$:

$$\mathcal{F}(f(t)) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t} dt = f(\omega) \quad (10)$$

And the inverse Fourier Transform of $f(\omega)$ is given by:

$$\mathcal{F}^{-1}(f(\omega)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(\omega) e^{i\omega t} d\omega = f(t) \quad (11)$$

Equations (10) and (11) are collectively known as the Fourier Transform pair.

1.1 The Discrete Fourier Transform

The discrete Fourier Transform is a mathematical technique used to analyse sampled data in the frequency domain, unlike the continuous Fourier Transform, which operates on continuous functions, the Discrete Fourier Transform is designed for data that is sampled at discrete points in time or space.

1.1.1 Derivation

We shall consider the Fourier Transform of a sampled time series, if the function $f(t)$ is sampled at N time points ΔT apart ;

$$f(t) = f(n\Delta T) \quad (12)$$

To make subsequent derivations easier, we require N to be an even number. The Fourier transform integral,

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt \quad (13)$$

This can be written as a summation ;

$$F(\omega) = \Delta T \sum_{n=0}^{N-1} f(n\Delta t)e^{-i\omega n\Delta t} \quad (14)$$

This transform is a continuous function of ω that we approximate using its values at discrete frequency points. Because sampling produces a spectrum that is periodic in angular frequency with period $\frac{2\pi}{\Delta T}$, or twice the Nyquist angular frequency (the Nyquist frequency (cycles per second) is the frequency whose cycle-length (or period) is twice the interval between samples, thus 0.5 cycle/sample.) ω_N , we divide this interval into N points as;

$$F(\omega) = F(K\Delta\omega) \text{ for } K = 0, 1, 2, \dots, N-1 \quad (15)$$

whereby;

$$\Delta\omega = \frac{2\omega_N}{N} = \frac{2\pi}{N\Delta T} = \frac{2\pi}{T} \quad (16)$$

where $T = N\Delta T$ is the total length of data in time. This sampled Fourier transform of a sampled time series is called the Discrete Fourier Transform (DFT);

$$f(K\Delta\omega) = \Delta t \sum_{n=0}^{N-1} f(n\Delta t)e^{-ik\Delta\omega n\Delta t} = \Delta t \sum_{n=0}^{N-1} f(n\Delta t)e^{-ikn\frac{2\pi}{N}} \quad (17)$$

The DFT gives values at angular frequencies

$$0, \Delta\omega, 2\Delta\omega, \dots, \frac{N}{2}\Delta\omega, \dots, (N-1)\Delta\omega \quad (18)$$

From equation (17) it is clear that the DFT and inverse DFT are given by;

$$F[k] = \sum_{n=0}^{N-1} f[n] \cdot e^{-j2\pi kn/N} \quad (19)$$

$$f[n] = \frac{1}{N} \sum_{k=0}^{N-1} [k] \cdot e^{j2\pi kn/N} \quad (20)$$

1.2 The Fast Fourier Transform

For these concepts to be useful, the transforms and inverse transforms must be evaluated on a computer. Moreover, it only makes sense to carry out filtering using Fourier transforms if the transform and inverse transform operations are relatively quick. It turns out that an elegant algorithm known as the Fast Fourier Transform (FFT) provides a fast way of carrying out the DFT.

1.2.1 Derivation

The underlying idea is that a simple method can be used to compute the transform of a series of points by splitting it in half. We take a series with N points,

$$f(n) = 0, 1, 2, \dots, N-1 \quad (21)$$

We can form 2 sub-series, one with even numbered points and one with odd numbered points;

$$a(n) = f(0), f(2), f(4), \dots = f(2n)$$

$$b(n) = f(1), f(3), f(5), \dots = f(2n+1)$$

$$n = 0, 1, 2, \dots, \frac{N}{2}$$

The DFTs of the sub-series are;

$$A(k) = \sum_{n=0}^{\frac{N}{2}-1} a(n) e^{-\frac{4\pi i k n}{N}} \quad (22)$$

$$B(k) = \sum_{n=0}^{\frac{N}{2}-1} b(n) e^{-\frac{4\pi i k n}{N}} \quad (23)$$

where k runs from 0 to $\frac{N}{2} - 1$ and the factor of 4 arises as the sub-series' lengths are $N/2$.

The DFT of the original series can then be written in terms of the sub-series;

$$F(k) = \sum_{n=0}^{N-1} f(n) e^{-\frac{2\pi i k n}{N}} \quad (24)$$

$$F(k) = \sum_{n=0}^{\frac{N}{2}-1} [a(n) e^{-\frac{2\pi i k (2n)}{N}} + b(n) e^{-\frac{2\pi i k (2n)}{N}}] \quad (25)$$

Equation (23) can be further simplified to;

$$F(k) = A(k) + e^{\frac{2\pi i k}{N}} B(k) \quad (26)$$

This gives the first $N/2$ points of $F(k)$, the second $N/2$ come from replacing k with $k+N/2$;

$$F(k + \frac{N}{2}) = A(k + \frac{N}{2}) + e^{\frac{2\pi i k}{N}} B(k + \frac{N}{2}) \quad (27)$$

Noting that because DFTs of the sub-series are periodic with period $N/2$, it follows that ;

$$A(k + \frac{N}{2}) = A(k) \quad (28)$$

and

$$B(k) = B(k + \frac{N}{2}) \quad (29)$$

for $k = 0, 1, 2, \dots, N/2-1$

The exponential can be written as;

$$e^{-2\pi i \frac{k + \frac{N}{2}}{N}} = e^{-\pi i} e^{-2\pi i \frac{k}{N}} = -e^{-2\pi i \frac{k}{N}} \quad (30)$$

the second half of the transform can be found from the first, using;

$$F(k + N/2) = A(k) - e^{-2\pi i \frac{k}{N}} B(k) \quad (31)$$

Substituting $W = e^{\frac{2\pi i}{N}}$ simplifies to ;

$$F(k) = A(k) + W^k B(k) \quad (32)$$

$$F(k + N/2) = A(k) - W^k B(k) \quad (33)$$

This method is called doubling a finding the transform of an N -point series from the transforms of its two $N/2$ -point sub-series. Doubling can be applied recursively, because we can find the transform of each $N/2$ -point series from that of two $N/4$ -point series, etc. Ultimately, a series of length $N = 2^n$ can be evaluated via $n = \log_2 N$ such stages. In the final stage, the transform of each 2-point series is found from two 1-point series, but the transform of a 1-point series is itself. Various methods can be used to further speed up operations. Thus, to obtain the FFT of a time series, we treat the data points as N 1-point series, use doubling to form $(N/2)$ 2-point series, and so on until the final N -point transform

We have assumed that the series length N is a power of 2. If this is not the case, a number of zeroes necessary to obtain a power of 2 can be added to the end of the time series.

To conclude this section, it is worth distinguishing between the DFT and the FFT. The DFT is the discrete approximation to the Fourier transform which has the periodic properties we have discussed. The FFT is a clever method for computing the DFT with much fewer operations.

2 Seismic Waves

Seismology is the study of the passage of elastic waves through the earth predominantly dealing with seismic waves. When an earthquake or explosion occurs part of the energy is released as elastic waves and transmitted through the earth. The waves are then detected by seismograms which measure, amplify and record the ground motion. The information can then be used for

- Earthquake monitoring
- Environmental and Geotechnical Studies
- Nuclear test monitoring
- Energy Exploration

among others. Seismic waves originate at the moving fracture line and travel through the earth's elastic media by reflection and refraction and thus reach the surfaces at different times and from different directions.

Seismic waves fall into two categories: body waves that travel through the body of the earth and surface waves that travel along the surface of the ground.

2.1 Body Waves

Travelling through the earth's interior, they arrive before surface waves emitted by an earthquake.

They are lower in amplitude and higher in frequency than surface waves. Body waves are of two types :

- Primary wave
- Secondary wave

2.1.1 Primary wave

It is the seismic wave of the highest frequency and consequently highest velocity. The wave is basically a compressional wave resulting from changes in the volume of the earth particles at the fracture in the lithosphere, as they are alternately condensed and expanded. It is transmitted through the earth at speeds 3 to 8 km/s, the exact speed depending on the nature of the strata through which the wave is travelling. The particles oscillate back and forth in the direction of the compressional wave, it is therefore a longitudinal wave.

2.1.2 Secondary Wave

The secondary wave is a transverse wave with the particles oscillating at right angles to the wave path and are second to arrive after an earthquake. The propagation of the S wave is limited to solids as they have a defined shear modulus. It is this property of S waves that led seismologists to the conclusion that the earth's outer core is fluid as S waves do not propagate through the earth's outer core. This is because fluids have no shear modulus.

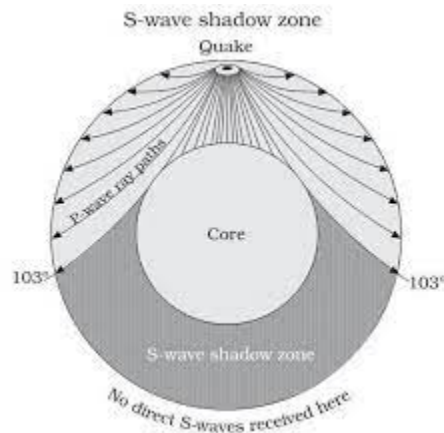


Figure 1: S wave propagation in the earth's interior

2.2 Surface Waves

Surface waves travel more slowly through Earth material at the planet's surface and are predominantly of lower frequency than body waves. They often have larger amplitudes than body waves and are thus the more destructive waves ($E \propto A^2$), where E is the energy of the wave and A is the amplitude. Surface waves have a slower decay rate than body waves and thus a longer duration which can subject structures to extended periods of stress.

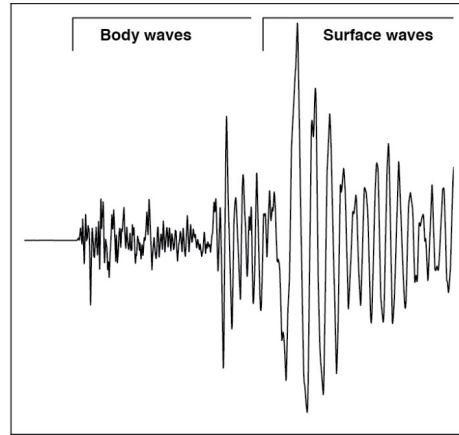


Figure 2: Variation in amplitude and frequency of body and surface waves

There are several types of surface wave, but the two most common varieties are Rayleigh waves and Love waves.

2.2.1 Rayleigh Waves

Rayleigh waves, also known as ground roll, spread through the ground as ripples, similar to rolling waves on the ocean. Like rolling ocean waves, Rayleigh waves move both vertically and horizontally in a vertical plane pointed in the direction in which the waves are travelling.

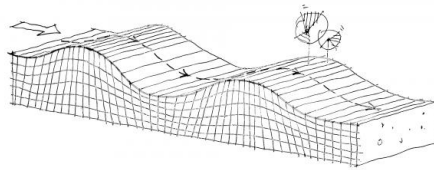


Figure 3: Rayleigh Wave

2.2.2 Love Waves

Love waves have the same motion as S-waves but without the vertical displacement. They move the ground from side to side in a horizontal plane but at right angles to the direction of propagation. Love waves are particularly damaging to the foundations of structures because of the horizontal ground motion they generate. Love also cause horizontal shearing of the ground. They are faster than Rayleigh waves but do not propagate through fluids.

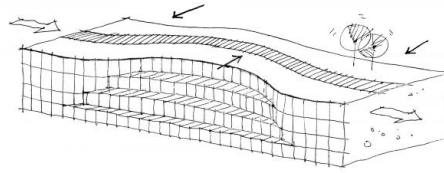


Figure 4: Love Wave

2.3 Simple Fourier Analysis in seismic waves

Waves are recorded on the seismogram as they are in the time domain and this makes it difficult for seismologists and engineers to distinguish the recorded waves. Luckily the Fourier Transform technique we previously discussed is a handy tool that enables to us to represent our signals in the frequency domain making it much easier to visually distinguish the recorded waves and perform other techniques such as filtering. An elementary example with synthetically generated waves is shown below.(in python programming language)

This is a recording as observed on a seismogram and it is clearly impossible to distinguish the individual components making up the wave.

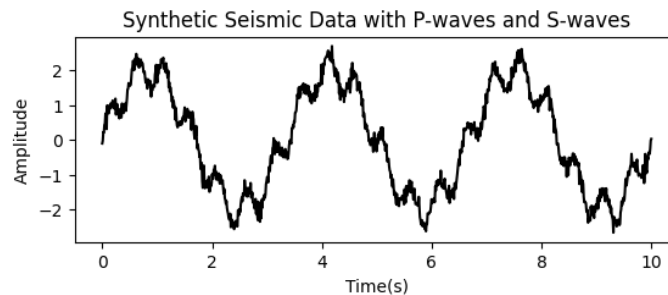


Figure 5: Time Domain

Taking the fourier transform of the signal and plotting it out gives ;

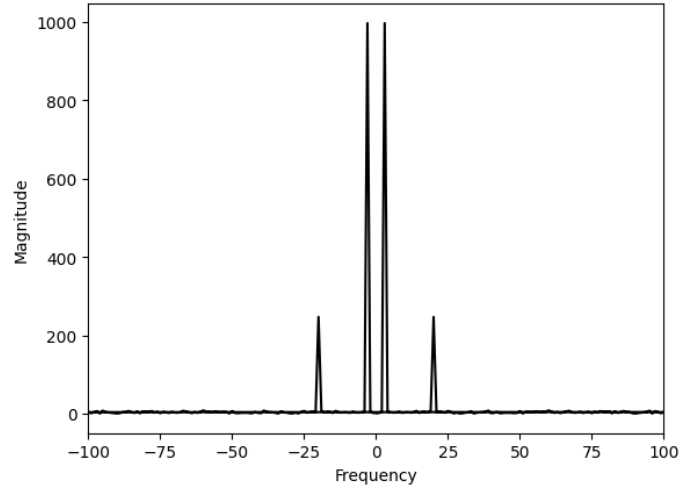


Figure 6: Frequency Domain

From the plot it is easy to distinguish the P from S waves, S waves have a low frequency but high magnitude whereas P waves have higher frequencies but lower magnitudes. The noise appears as small peaks in the relatively flat background. Working in the frequency domain makes it easier to do subsequent analysis such as filtering and spectral analysis.

Performing a power spectral density analysis:

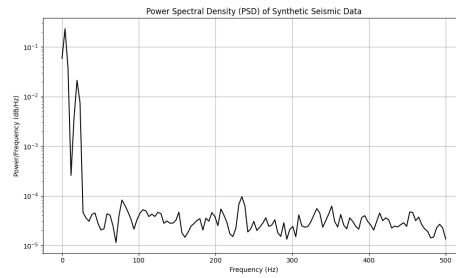


Figure 7: Wave Power

3 Fourier Transform in Dynamics of Structures

In this section we shall discuss single degree of freedom systems with forced periodic loading.
Consider the following

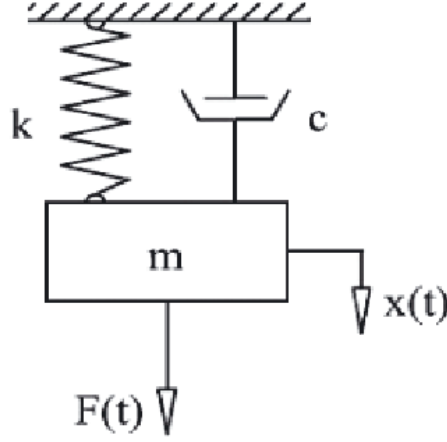


Figure 8: Forced vibration with force $f(t)$

where $F(t)$ represents the external harmonic load, k is the spring constant, m the mass, $x(t)$ is the position function and c the damping.
The dynamic equilibrium equation of this system can be given by

$$F(t) = m\ddot{x} + c\dot{x} + kx \quad (34)$$

and $P(t)$ is given by ;

$$P(t) = P_o \sin(\Omega t) \quad (35)$$

where Ω is the frequency of the applied load and ω is the natural frequency of the system given by $\sqrt{\frac{k}{m}}$

We shall solve equation (34) in the time domain, and more importantly, in the frequency domain as well.

3.1 Time domain solution

The solution of equation (34) will consist of two parts the particular solution $X_p(t)$ and the complementary solution $X_c(t)$. The complete solution will then be the algebraic sum of the two,

$$X(t) = X_p(t) + X_c(t) \quad (36)$$

The complementary solution is obtained by letting $F(t) = 0$, making equation (34) a homogeneous differential equation with constant coefficients. It is a standard well known equation in mathematics whose solution is ;

$$X_c = e^{-\xi\omega t}(A\sin(\omega_D t) + B\cos(\omega_D t)) \quad (37)$$

where ω_D represents the damped natural frequency of the system. The particular solution to equation (36) is given by

$$X_p = X\sin(\omega t) \quad (38)$$

substituting equation 38 into equation 34 yields;

$$X(p) = \frac{P_o}{k} \left[\frac{1}{\sqrt{(1-\beta^2)^2 + (2\beta\xi)^2}} \right] \cos(\Omega t - \theta) \quad (39)$$

The complementary solution (38) dies out with time, and is the transient response which exponentially dies out with time and can hence be ignored , the total response can then be described by equation (39).

The solution represents the structure's displacement at all points in time. The coefficient $\frac{P_0}{k}$ represents the static displacement whereas $\frac{1}{\sqrt{(1-\beta^2)^2 + (2\beta\xi)^2}}$ represents the dynamic magnification factor, which is the ratio of dynamic to static deflection. At resonance , the forcing frequency is equal to the natural frequency of the system($\omega = \Omega$ and $\beta = 1$) and this leads to excessive displacements. It can be shown mathematically that substituting $\beta = 1$ in the solutions yields:

$$U(t) = \frac{P_0}{2\xi k} \quad (40)$$

3.2 Frequency domain solution

Equation 34 however can be solved in the frequency domain;

The input force is $P_o \sin(\Omega t)$ whereas the steady state response is given by $X \sin(\Omega t - \phi)$ and ϕ is the phase lag

In one approach, the steady-state response may be determined by substituting these expressions into the equation of motion and then equating sine and cosine terms using trigonometric expansion. However, an alternative approach uses complex algebra and will be adopted since it is more powerful and commonly used. In this approach, the force and response are rewritten, somewhat artificially, in a complex notation such that:

$$f(t) = F e^{i\Omega t} = F \cos(\Omega t) + i \sin(\Omega t) \quad (41)$$

$$x(t) = X e^{i(\Omega t - \phi)} = \hat{X} \cos(\Omega t) + i \hat{X} \sin(\Omega t) \quad (42)$$

Here, the excitation is rewritten in terms of sine and cosine components and the phase lag is Only the imaginary part of the excitation embedded in a new complex amplitude quantity X . and response will be used for sine excitation; an alternative way of viewing this approach is that the solutions for both the sine and cosine excitation will be found simultaneously.

Noting that $\dot{x} = i\Omega \hat{X} e^{i\omega t}$ and $\ddot{x} = -\Omega^2 \hat{X} e^{i\Omega t}$ and cancelling the exponential term, then:

$$F = (-\Omega^2 m + i\Omega c + K) \hat{X} \quad (43)$$

Thus the complex response amplitude may be solved from this algebraic equation so that:

$$\hat{X} = X e^{i\phi} = \frac{F}{k - \Omega^2 m + i\Omega c} \quad (44)$$

and equating real and imaginary parts from the two sides of this equation yields the amplitude and phase as

$$X = \frac{F}{\sqrt{(k - \Omega^2 m)^2 + (\Omega c)^2}} \quad (45)$$

and

$$\phi = \arctan\left(\frac{\Omega c}{k - \Omega^2 m}\right) \quad (46)$$

Hence, the steady-state time response may be calculated using X, ϕ from this equation.

An alternative way of writing Equation (45) is

$$H_D(\Omega) = \frac{\hat{X}}{F} = \frac{1}{k - \Omega^2 m + i\Omega c} \quad (47)$$

And in non-dimensional form, where $r = \frac{\Omega}{\omega}$

$$H_D(\Omega) = \frac{1/k}{1 - r^2 + 2i\zeta r} \quad (48)$$

Here $H_D(\Omega)$ is known as the displacement Frequency Response Function (FRF) of the system and is a system property; it dictates how the system behaves under harmonic excitation at any frequency. The equivalent velocity and acceleration FRFs are given by :

$$H_V = i\Omega H_D \quad (49)$$

$$H_A = -\Omega^2 H_D \quad (50)$$

since multiplication by $i\Omega$ in the frequency domain is equivalent to differentiation in the time domain.

The quantity $kH_D(\Omega)$ from Equation (48) is a non-dimensional expression, or dynamic magnification, relating the dynamic amplitude to the static deformation for several damping ratios. The well-known ‘resonance’ phenomenon is shown in Figure 9 by the amplitude peak that occurs when the excitation frequency Ω is at the ‘resonance’ frequency, close in value to the undamped natural frequency ω ; the phase changes rapidly in this region, passing through 90° at resonance. Note that the resonance peak increases in amplitude as the damping ratio reduces and that the dynamic magnification (approximately $\frac{1}{2\zeta}$) can be extremely large.

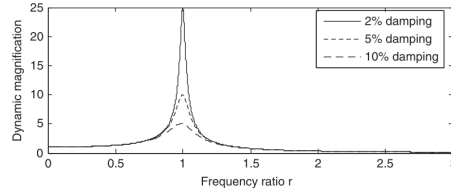


Figure 9: Displacement Frequency Response Function for an SDoF system

In practice, the definition of the FRF may be extended to cover a more general excitation by employing the Fourier Transform (FT), so that

$$H(\Omega) = \frac{X(\Omega)}{F(\Omega)} = \frac{\mathcal{F}(x(t))}{\mathcal{F}(f(t))} \quad (51)$$

The Fourier Transform $X(\omega)$ is a complex function of frequency (i.e. a spectrum) whose real and imaginary parts define the magnitude of the components of $\cos()$ and $-\sin(\omega t)$ in the signal. $x(t)$. The units of $X(\omega)$, $F(\omega)$ in this definition are typically ms and Ns and the units of $H(\omega)$ are m/N. The Inverse Fourier Transform, not defined here, allows the frequency function to be transformed back into the time domain.

It may be seen that rearranging equation (51) leads to:

$$X(\Omega) = H(\Omega)F(\Omega) \quad (52)$$

A useful feature of Equation (52) is that it may be used to determine the response of a system, given the excitation time history, by taking a frequency domain route. Thus the response $x(t)$ of a linear system to a finite length transient excitation input $f(t)$ may be found by the following procedure:

1. Fourier Transform $f(t)$ to find $F(\Omega)$.
2. Determine the FRF $H(\Omega)$ for the system.
3. Multiply the FRF by $F(\Omega)$ using Equation (52) to determine $X()$
4. Inverse Fourier Transform $X(\Omega)$ to find $x(t)$

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